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Questions

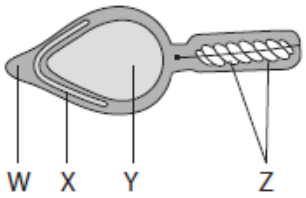
Q1.

The phenotype of a rabbit is affected by its genotype.

Rabbit sperm cells contain genetic information that will become part of the genotype of the offspring.

The diagram shows part of a rabbit sperm cell.

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(a) Which labelled structure contains the genetic information used in the synthesis of acrosomal enzymes?

(1)

☒ **A** W

☒ **B** X

☒ **C** Y

☒ **D** Z

(b) A fertilised rabbit egg cell will divide by mitosis to form a solid ball of genetically identical cells.

Name this solid ball of cells.

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(1)

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(c) The genetically identical cells in a developing rabbit embryo will continue to divide by mitosis.

Some of the cells produced can undergo epigenetic modification as they develop into specialised cells.

(i) Epigenetic modification can alter the activation of certain genes.

Describe **one** form of epigenetic modification that could occur in the rabbit cells.

(2)

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(ii) Explain how cells in the rabbit embryo become specialised after epigenetic modification.

(3)

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* (d) The table shows information about three types of rabbit.

Type of rabbit	Black 	Himalayan 	Albino 
Colour of fur	Solid black all over	White fur with black ears, nose, paws and tail	White all over
Allele combination(s)	CC, Cc^{ch}, Cc^{a}	$c^{h}c^{h}$, $c^{h}c^{a}$	$c^{a}c^{a}$

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Enzymes produced by the rabbits control the production of pigments in the fur.

One type of rabbit produces an enzyme that is inactive at the core body temperature of the rabbit.

Discuss how genes and the environment interact to produce the fur colour of these three types of rabbit.

Use the information in the question to support your answer.

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(Total for question = 13 marks)

Q2.

Egg cells vary in size.

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(a) The table shows the size of egg cells from three species of animal.

Species	Diameter of egg cell / µm
Orb weaver spider	10
Human	150
Blue poison dart frog	1600

(i) Which of the following shows the diameter, in standard form, of the egg cell of a blue poison dart frog?

(1)

☒ **A** 1.6×10^{-2} µm

☒ **B** 1.6×10^{2} µm

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☒ **C** 1.6×10^{-3} µm

☒ **D** 1.6×10^{3} µm

(ii) The egg cells of humans and blue poison dart frogs are approximately spherical.

The volume of a sphere (V) is calculated using the formula

$$V = \frac{4}{3}\pi r^3$$

The volume of the human egg cell is 1.8×10^{6} µm^3.

Calculate how many times larger the volume of the blue poison dart frog egg cell is than the volume of a human egg cell.

Give your answer to two significant figures.

(3)

Answer .....

(b) An egg cell and a skin cell were removed from a female human.

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Both cells contained a nucleus.

The skin cell contained 46 chromosomes.

Explain the differences between the genetic material from these two cells.

(4)

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(Total for question = 8 marks)

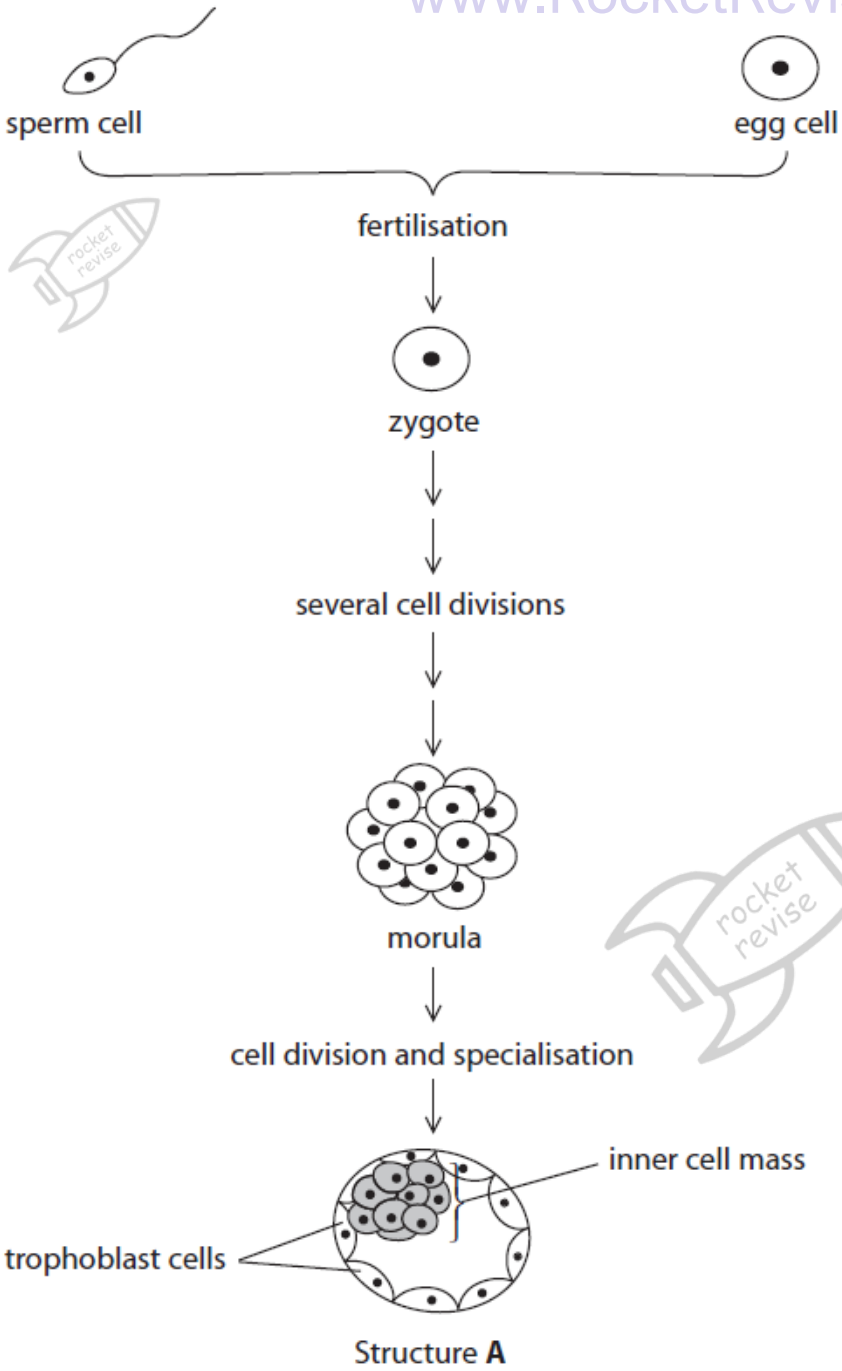
Q3.

The diagram shows three structures that form after the fertilisation of an egg cell.

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(a) Explain why the cells in the morula are genetically identical.

(2)

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(b) Structure **A** can be used as a source of pluripotent stem cells.

(i) Name structure **A**.

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(1)

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(ii) Stem cells can be used as a medical treatment when a patient has had a severe heart attack.

A severe heart attack can cause the death of heart muscle cells.

The medical treatment involves injecting stem cells into damaged areas of the heart muscle.

Suggest why injecting pluripotent stem cells may benefit a person who has had a severe heart attack.

(2)

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(Total for question = 5 marks)

Q4.

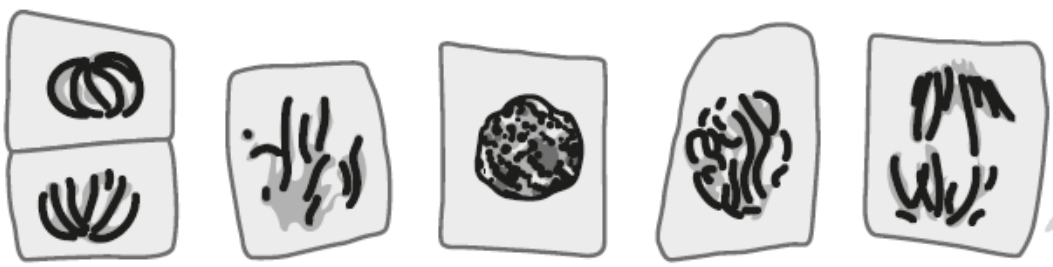
Cells undergo cell division.

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(a) The diagrams show cells from a plant.

The cells are in different stages of division.

These cells do not show the correct order for the process of division by mitosis.



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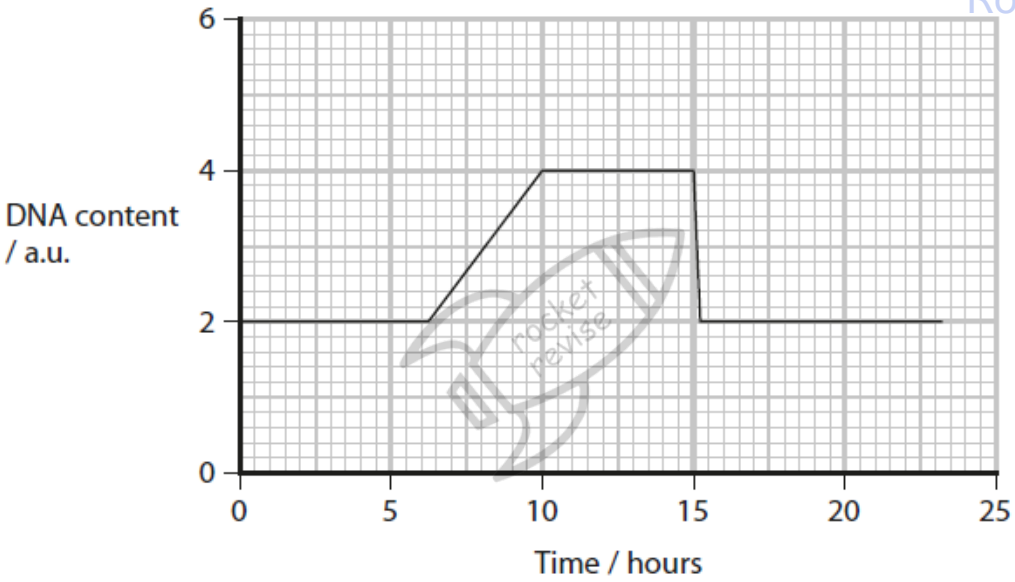
Which of the following shows the correct order for the process of division by mitosis?

(1)

	1st	2nd	3rd	4th	5th
☒ **A**	A	E	B	D	C
☒ **B**	A	D	C	E	B
☒ **C**	C	D	B	E	A
☒ **D**	C	B	E	D	A

(b) The graph shows the changes in the DNA content of a cell during one cell cycle.

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(i) Explain the changes in the DNA content as shown by this graph.

(4)

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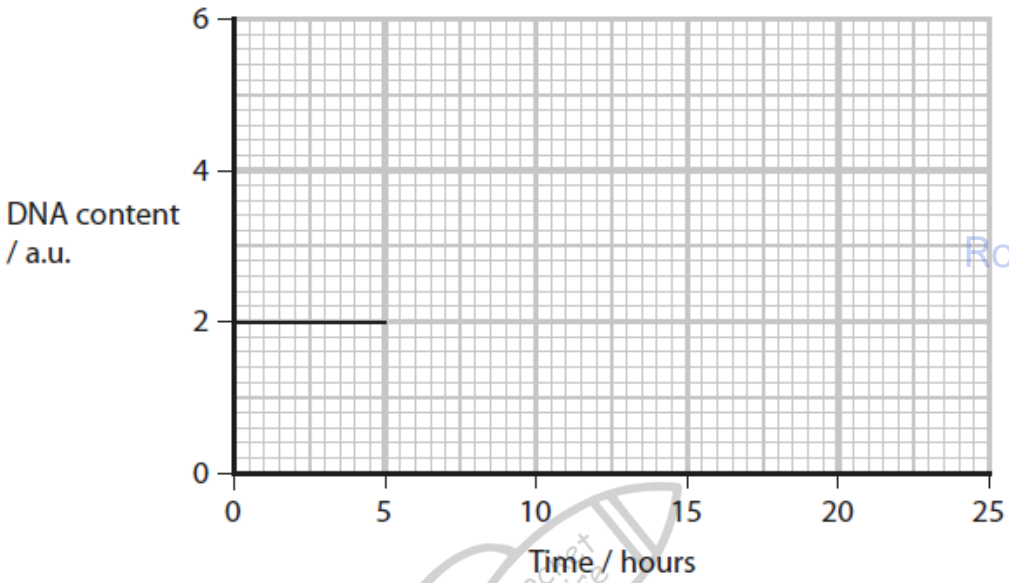
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(ii) Complete the graph to show how the DNA content would change if the cell had undergone meiosis to form gametes.

(2)



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(Total for question = 7 marks)

Q5.

The photograph shows a species of harvest mouse native to countries in Asia.



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(Source: © Mark Bridger/Shutterstock)

(a) The body of this species of harvest mouse is 55 mm to 75 mm long and its tail is 50 mm to 75 mm long.

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The body of the Indian flat-haired mouse is 10 cm to 12 cm long and its tail is 7 cm to 8 cm long.

Calculate the percentage difference in **minimum** total length of the Indian flat-haired mouse compared with the **minimum** total length of the harvest mouse.

Give your answer to **two** significant figures.

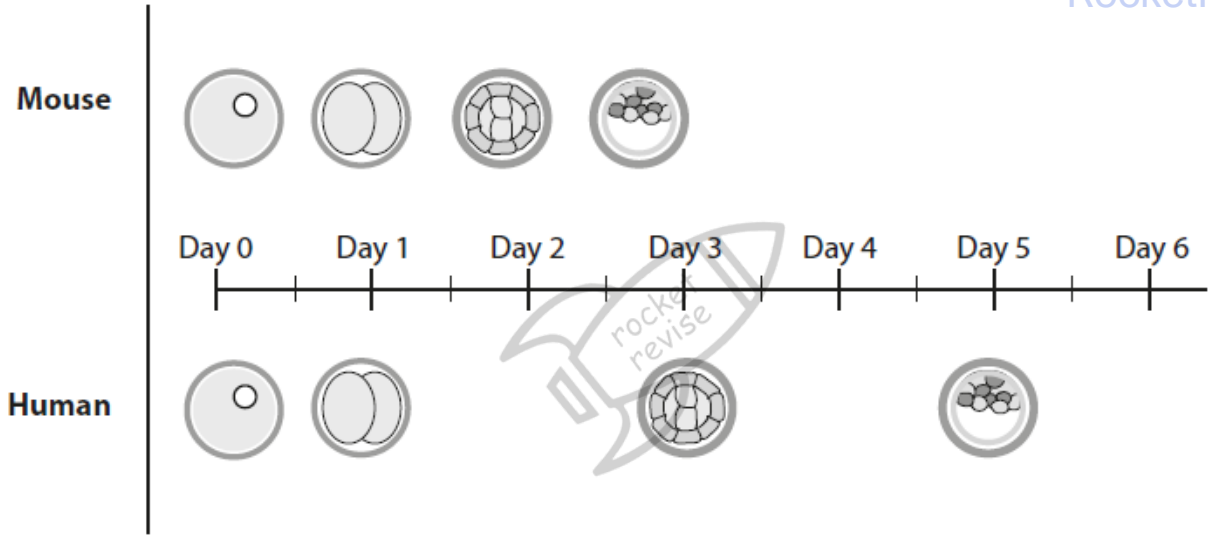
(2)

Answer ..... %

(b) The diagram shows the timeline for the structures formed after fertilisation in a mouse and in a human.

A fertilised egg cell is called a zygote.

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(i) As mouse egg cells develop they accumulate small spherical lipid droplets.

These lipid droplets have a mean radius of 88 μm.

The volume of a sphere (V) is calculated using the formula

$$V = \frac{4}{3}\pi r^3$$

Which of the following gives the mean volume of a lipid droplet in **mm^3**?

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(1)

☒ **A** 2.85×10^{-6}

☒ **B** 2.85×10^{-3}

☒ **C** 2.85×10^{3}

☒ **D** 2.85×10^{6}

(ii) Draw a circle around the part of the diagram that is a mouse morula. Label the circle 'morula'.

(1)

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(iii) Describe the events that occur after a sperm cell enters an egg cell, until a zygote is formed.

(3)

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(c) Analysis of some mouse egg cells showed that the lipid droplets were located within 10 nm of mitochondria.

Suggest why mitochondria were located within 10 nm of the lipid droplets.

(3)

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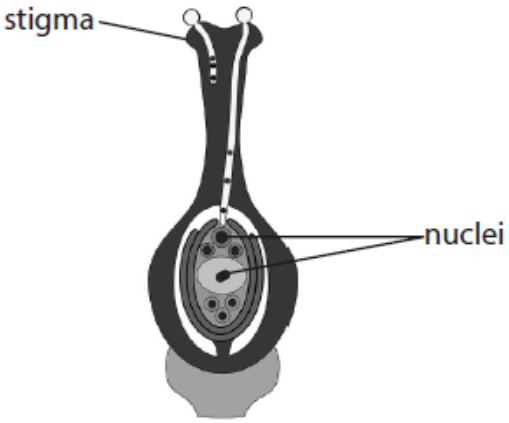
(Total for question = 10 marks)

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Q6.

The diagram shows the nuclei in an ovule of a flowering plant.



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(Source: © Kazakova Maryia/Shutterstock)

(a) Double fertilisation takes place in the ovules.

Describe the events that occur from the time a pollen grain lands on the stigma to the production of a triploid endosperm nucleus and a zygote.

Use the information in the diagram to support your answer.

(5)

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(b) The photograph shows flowers from a silver trumpet tree in Thailand.



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(Source: © Picture/Shutterstock)

Flowering plants such as the silver trumpet tree contain several ovules.

Explain how the silver trumpet tree produces seeds that are genetically different from each other.

(4)

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(Total for question = 9 marks)

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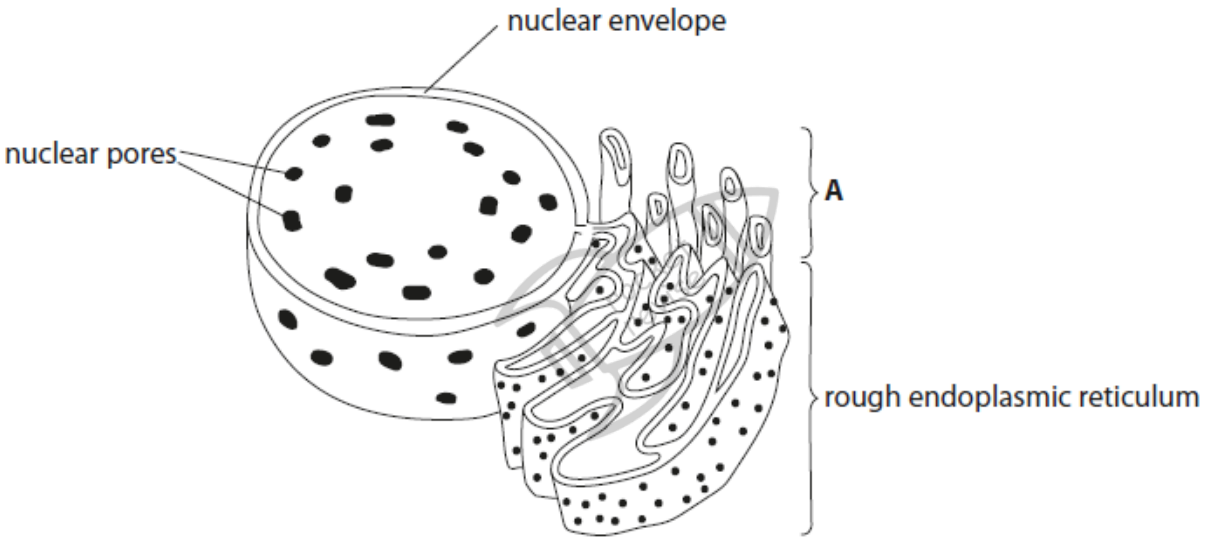
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Q7.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Eukaryotic cells contain cell organelles.

(a) The diagram shows part of a eukaryotic cell, as drawn by a student.
The nuclear envelope, nuclear pores and rough endoplasmic reticulum are labelled.



(i) Which of the following is a correct statement about rough endoplasmic reticulum?

(1)

☒ **A** it has 70S (small) ribosomes

☒ **B** it has 80S (large) ribosomes

☒ **C** it forms vesicles surrounding phospholipids only

☒ **D** it forms vesicles surrounding polysaccharides only

(ii) Name a substance synthesised by the organelle labelled **A** in the diagram.

(1)

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(iii) Name **two** structures that could be present inside the nucleus in the diagram.

(2)

1 .....

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(b) After cell division, the plant cells produced increase in size.
Explain how these plant cells increase in size.

(3)

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(Total for question = 7 marks)

Q8.

Gametes are specialised for their function.

Egyptian female desert ants (*Cataglyphis savignyi*) mate with more than one male.

A male desert ant has an adaptation to improve the chances of his sperm fertilising a female's egg cell.

The sperm are released in bundles with the heads held together by a sticky protein.

The diagram shows a bundle of desert ant sperm.

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(a) Suggest how the sticky proteins would have been secreted.

(2)

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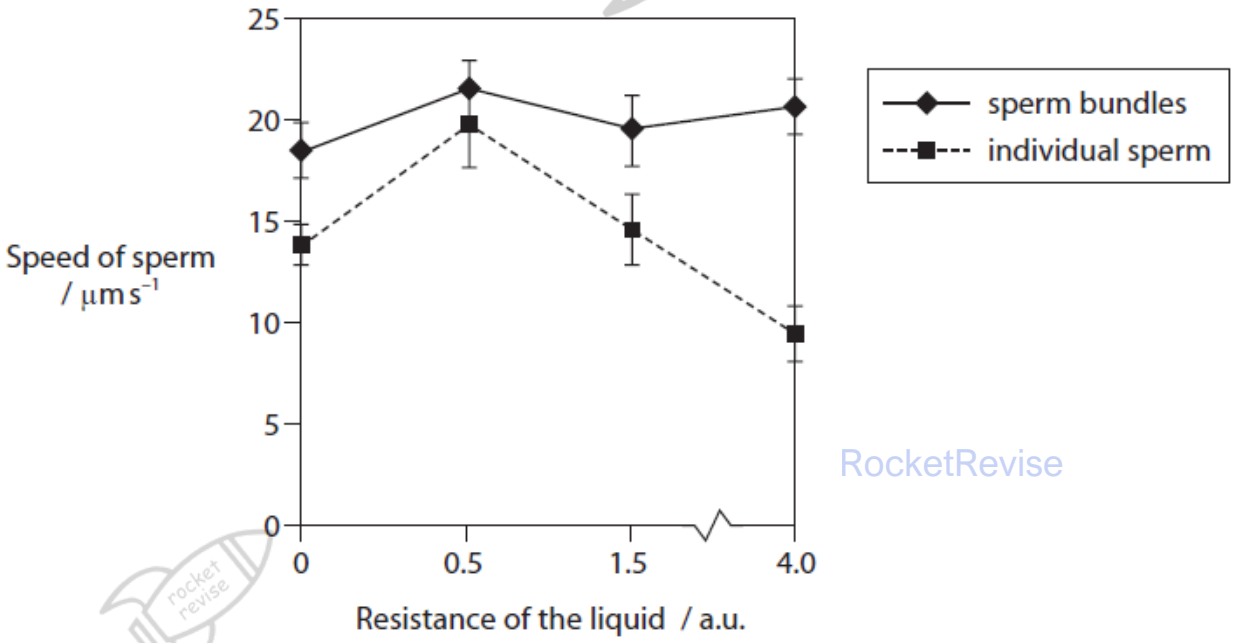
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(b) The graph shows the speed of individual sperm compared with bundles of sperm in liquids of different resistance.



Comment on the results of this investigation.

(4)

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(c) The photograph shows an Amur leopard found in southeastern Russia and northern China.

In-vitro fertilisation (IVF) techniques are being developed to help conserve this endangered species of leopard.



(Source: © NaturaLight / Alamy Stock Photo)

These techniques are being developed using mice.

IVF involves the removal of egg cells from the female mice. The zona pellucida can be damaged during this process.

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The effect of damaging the zona pellucida was investigated.

Three hundred mouse egg cells were distributed equally in three Petri dishes containing nutrient solutions.

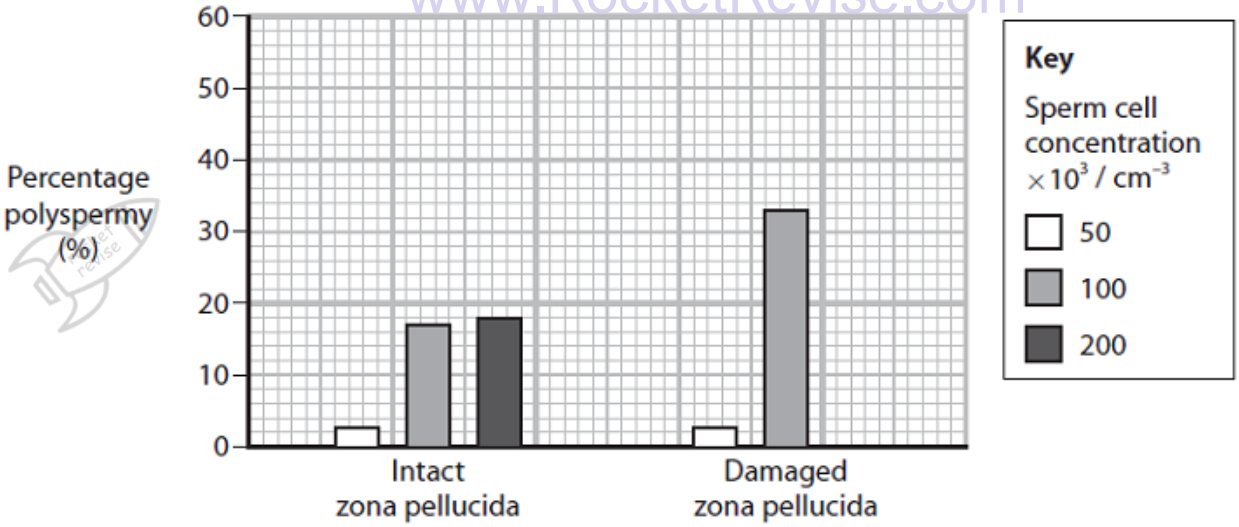
Samples of sperm of different concentrations were added to each Petri dish. The volume of each sample added was the same.

The experiment was repeated for egg cells which had their zona pellucida damaged.

The graph shows the percentage of egg cells fertilised by more than one sperm (percentage polyspermy).

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(i) When the highest sperm concentration was added to the egg cells with damaged zona pellucida, the percentage polyspermy was 52%.

Plot this result on the graph.

(1)

(ii) Explain the results of this investigation.

(4)

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(Total for question = 11 marks)

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Q9.

Fertilisation in mammals, such as the African northern white rhino, results in the fusion of gamete nuclei.

The photograph shows the last two remaining northern white rhinos.

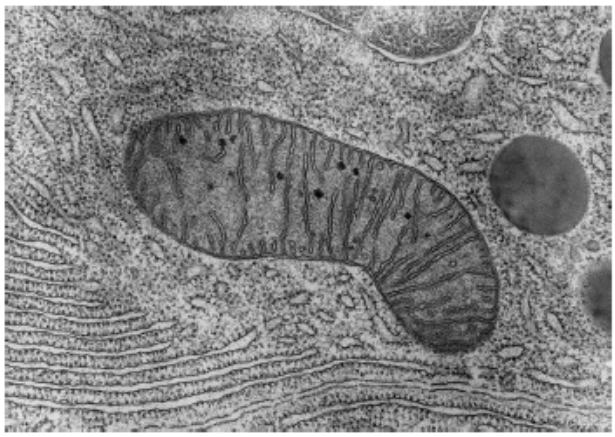
Both of these rhinos are female.

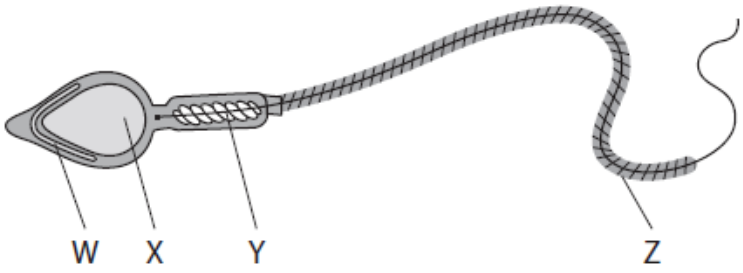


(Source: © Xinhua/Alamy Stock Photo)

Sperm and tissue samples were taken from the last few male northern white rhinos before they died. The sperm samples were frozen.

(a) The photograph shows an organelle taken from a sperm cell, as seen using an electron microscope. The diagram shows a sperm cell.





(Source: © Science History Images/Alamy Stock Photo)

In which part of this sperm cell would this organelle be found?

(1)

☒ **A** W

☒ **B** X

☒ **C** Y

☒ **D** Z

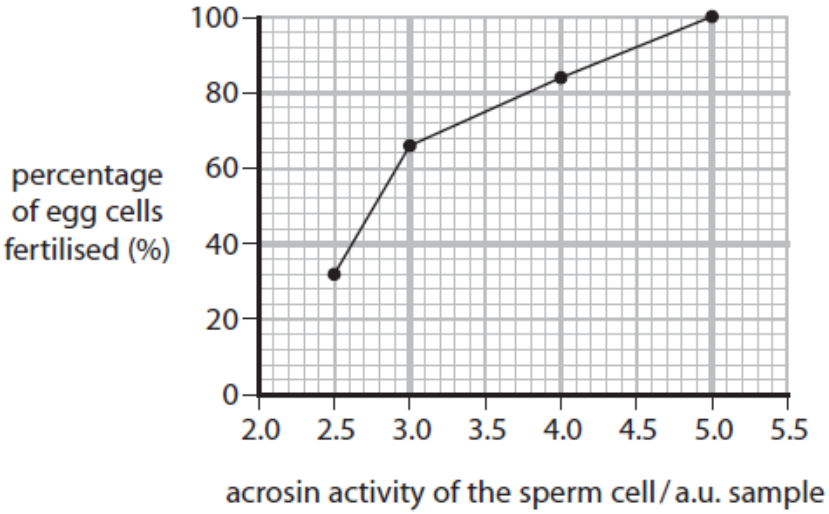
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(b) Sperm cells contain a digestive enzyme called acrosin.

The effect of acrosin activity of samples of sperm cells on the percentage of egg cells fertilised was investigated.

The graph shows the results of this investigation.



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(i) Describe the role of **one** organelle involved in the production of this enzyme.

Organelle

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Role

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(ii) The investigation used 2500 egg cells for each sample of sperm cells.

Calculate the number of egg cells that would have been fertilised by sperm cells with 2.5 a.u. acrosin activity.

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(1)

Answer .....

(iii) Explain why a higher activity of the digestive enzyme acrosin resulted in a higher percentage of egg cells being fertilised.

(4)

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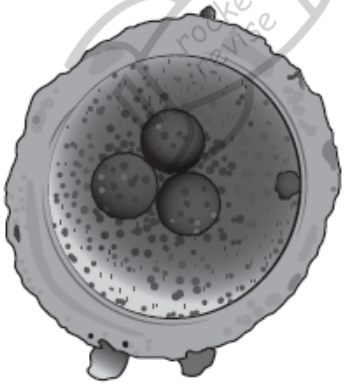
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(c) In vitro fertilisation (IVF) has been used to fertilise northern white rhino egg cells using the frozen sperm samples.

The zona pellucida of some egg cells were damaged when they were removed from the ovaries of the females.

The drawing shows a fertilised egg cell with three nuclei.



magnification ×200

(i) State how to calculate the actual size of this egg cell.

(1)

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(ii) Suggest why this fertilised egg cell has three nuclei.

(3)

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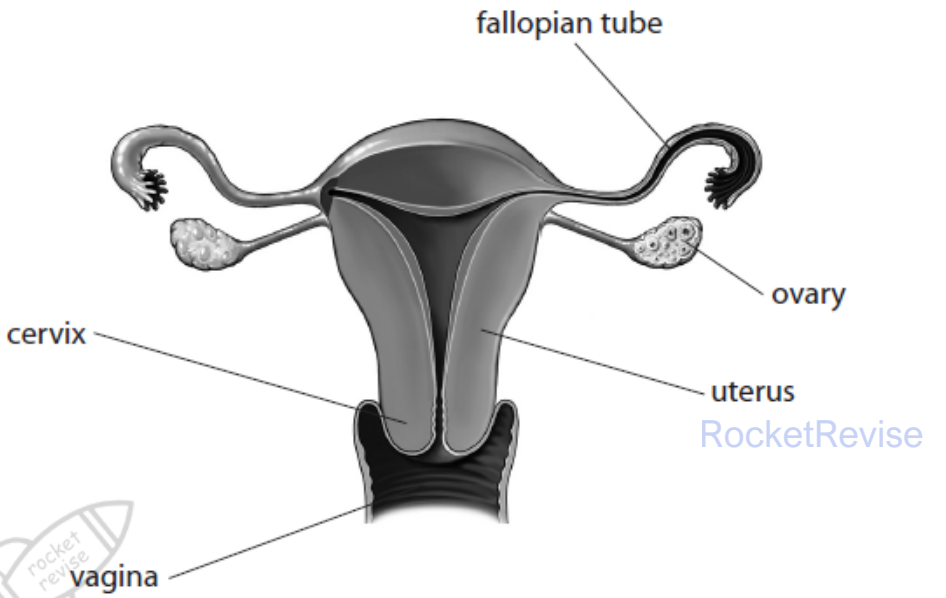
(Total for question = 12 marks)

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Q10.

Answer the questions with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The diagram shows an organ system of a human female.



(Source: © Nucleus Medical Media Inc / Alamy Stock Photo)

(a) State what is meant by the term **organ system**.

(1)

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(b) Gametes are produced in one of the tissues in the ovary.

(i) State what is meant by the term **tissue**.

(1)

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(ii) Gametes are specialised for their functions.

Complete the table to show the correct statement for each structure found in gametes.

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(4)

Structure	Propels male gamete towards female gamete	Modified by the action of cortical granules	Produces ATP by respiration	Contains linear DNA
flagellum	☒ A	☒ B	☒ C	☒ D
mitochondria	☒ A	☒ B	☒ C	☒ D
nucleus	☒ A	☒ B	☒ C	☒ D
zona pellucida	☒ A	☒ B	☒ C	☒ D

(Total for question = 6 marks)

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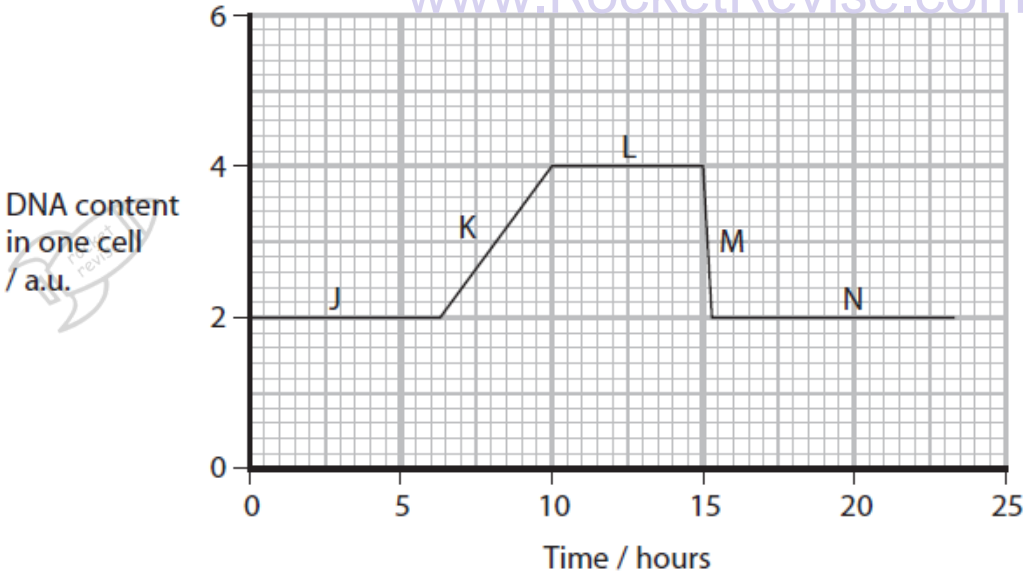
Q11.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The graph shows the changes in the DNA content of one cell during one cell cycle.

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(a) (i) In which part of the graph is the DNA being replicated?

(1)

☒ **A** J

☒ **B** K

☒ **C** L

☒ **D** M

(ii) In which part of the graph would the condensation of chromosomes be completed?

(1)

☒ **A** J

☒ **B** L

☒ **C** M

☒ **D** N

(b) Centrioles are involved in spindle formation.

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(i) Name the stage of mitosis when spindle fibres begin to form.

(1)

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(ii) Explain the role of the spindle in mitosis.

(2)

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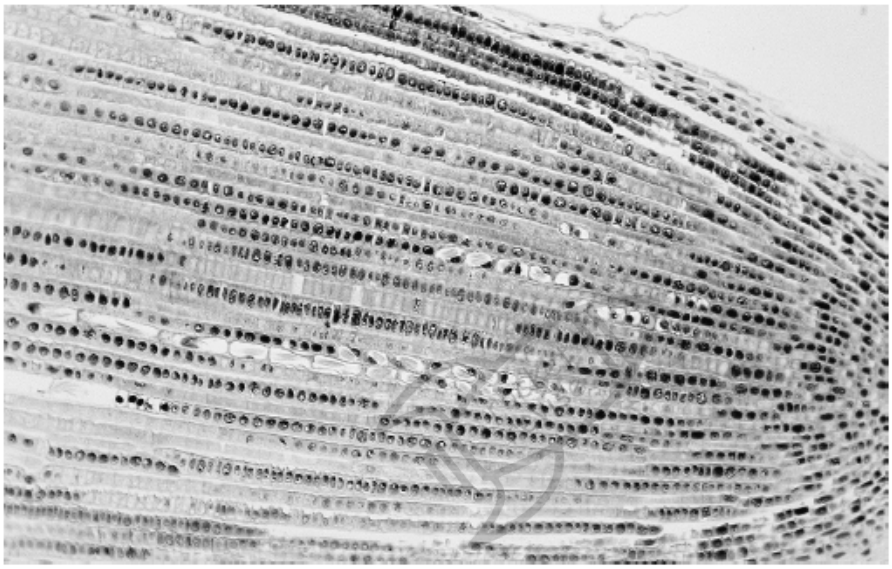
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(iii) Draw a plant cell undergoing cell division, after mitosis has just finished.

(2)

(c) The photograph shows a section through a root tip, as seen using a microscope.



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(Source: © agefotostock / Alamy Stock Photo)

Describe how to determine the mitotic index of this tissue.

(2)

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(Total for question = 9 marks)

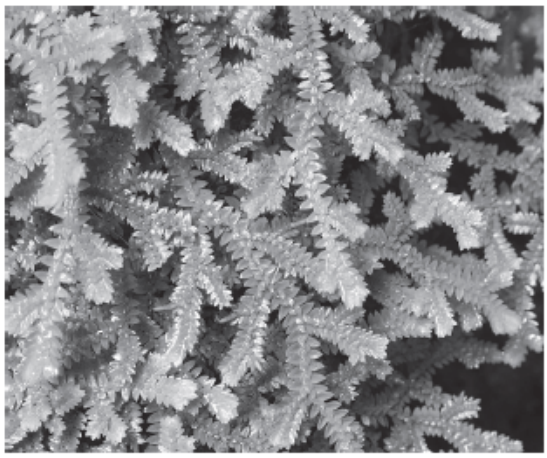
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Q12.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows a type of plant called a liverwort. This plant grows in Costa Rica.

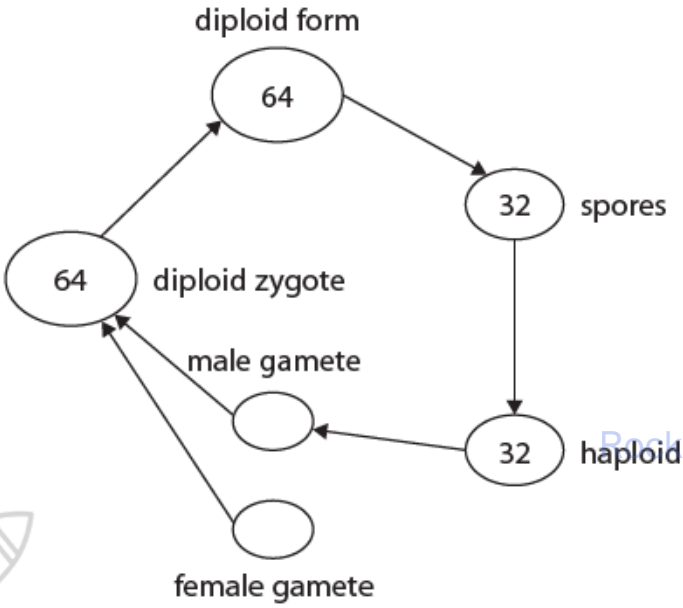


(Source: © Steve Taylor ARPS/Alamy Stock Photo)

(a) Liverworts reproduce sexually with gametes and asexually with spores.

The diagram shows the life cycle of a liverwort, which exists in two forms.

The diploid form contains 64 chromosomes and the haploid form contains 32 chromosomes.



(i) Complete the diagram to show the number of chromosomes in the female and male gametes.

(1)

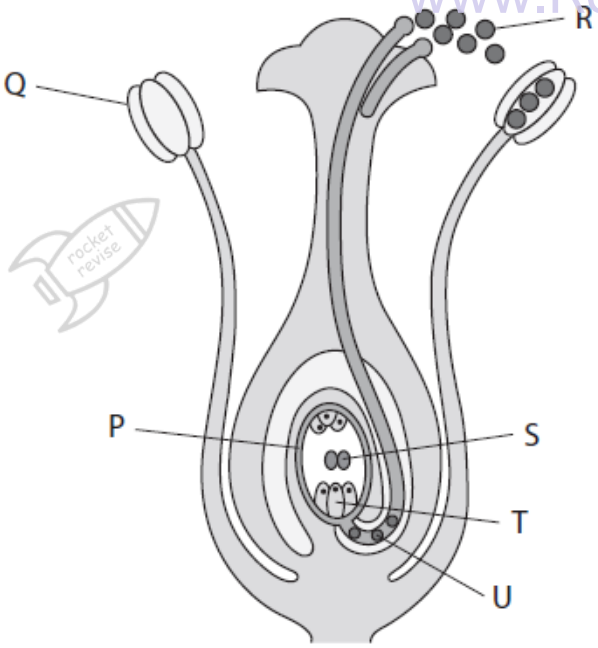
(ii) Label the part of the diagram where meiosis would have occurred.

(1)

(b) The diagram shows the nuclei in an ovule and a pollen tube in a flowering plant.

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(i) Which letter labels an ovule?

(1)

☒ **A** P

☒ **B** Q

☒ **C** S

☒ **D** T

(ii) Which letter labels an egg cell (female gamete)?

(1)

☒ **A** P

☒ **B** S

☒ **C** T

☒ **D** U

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(iii) Which letter labels a male nucleus?

(1)

☒ **A** R

☒ **B** S

☒ **C** T

☒ **D** U

(iv) Which letter labels a polar nucleus?

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(1)

☒ **A** R

☒ **B** S

☒ **C** T

☒ **D** U

(v) Describe the function of the tube nucleus.

(2)

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(Total for question = 8 marks)

Q13.

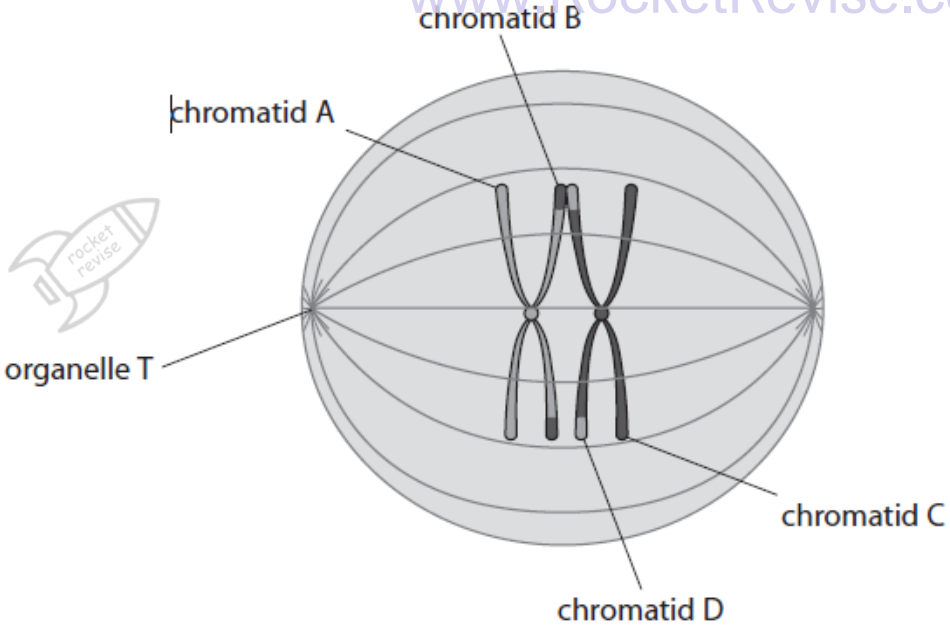
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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The diagram shows part of a cell at the start of metaphase I of meiosis.

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(a) Which is organelle T?

(1)

- ☒ **A** centriole
- ☒ **B** lysosome
- ☒ **C** nucleolus
- ☒ **D** ribosome

(b) Explain why it is possible for chromatid A to be genetically different from chromatid B.

(2)

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(c) A student was asked to draw the four haploid cells produced by this cell in metaphase I of meiosis.

The diagram shows the incomplete drawings produced by this student.

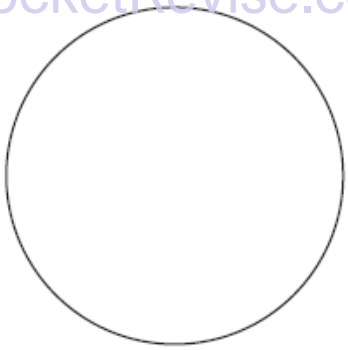
Complete the drawings to show the **four haploid** cells produced.

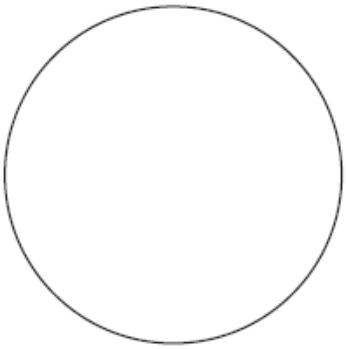
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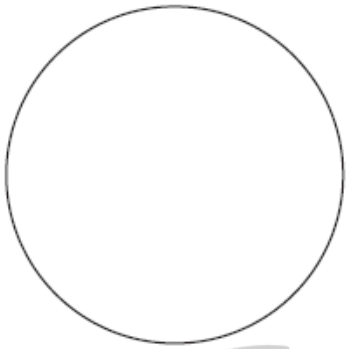
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(Total for question = 5 marks)

Q14.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Meiosis results in genetic variation in gametes.

(a) (i) Explain how meiosis causes this genetic variation.

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(3)

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(ii) Give **two** differences in the arrangement of the DNA in a cell at the beginning of interphase and at the end of prophase I.

(2)

1 .....

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2 .....

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(b) The DNA content of two cells was compared.

Cell A was undergoing mitosis. The mass of DNA in the cell in prophase was 12 pg.

Cell B was undergoing meiosis. The mass of DNA in the cell in prophase I was 12 pg.

What would be the mass of DNA in each daughter cell produced at the end of mitosis and at the end of meiosis?

(1)

	Mass of DNA in a daughter cell produced from cell A at the end of mitosis / pg	Mass of DNA in a daughter cell produced from cell B at the end of meiosis / pg
☐ **A**	3	6
☐ **B**	6	3
☐ **C**	6	12
☐ **D**	12	6

*(c) Plants can produce seeds from either cross-fertilisation or self-fertilisation.

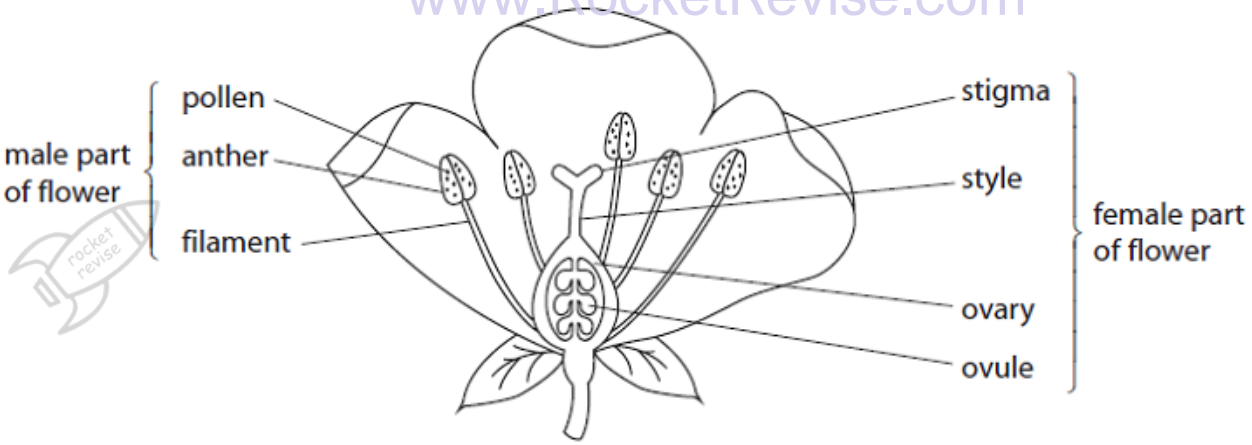
Self-fertilisation occurs when male nuclei from a flower fertilises an egg cell in a flower on the same plant.

Cross-fertilisation occurs when male nuclei from a flower on one plant fertilises an egg cell in a flower on a different plant of the same species.

The image shows the female and male parts of a flower.

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Some plants have developed strategies to prevent self-fertilisation.

- Date palm plants have only male flowers or only female flowers.
- Some orchids release chemicals that prevent the germination of pollen produced by flowers of the same plant.

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Some plants have developed strategies to produce seeds if cross-fertilisation has not occurred.

- The fireweed plant species from North America will self-fertilise if cross-fertilisation has not occurred.
- Unfertilised egg cells of a dandelion plant will develop into a seed.

Discuss the advantages and disadvantages of the strategies shown by these plants.

(6)

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(Total for question = 12 marks)

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Q15.

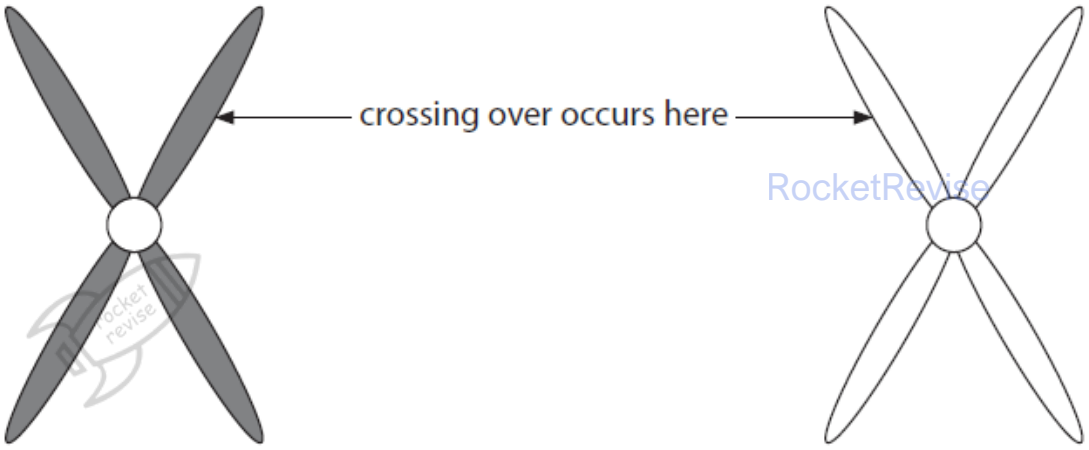
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Meiosis and mitosis are involved in cell division.

(a) Meiosis produces gametes that are genetically different.

Crossing over is important to increase genetic variation.

The diagram shows one pair of homologous chromosomes during early meiosis.

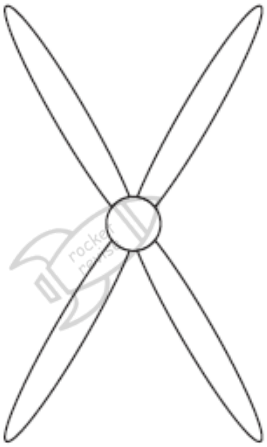


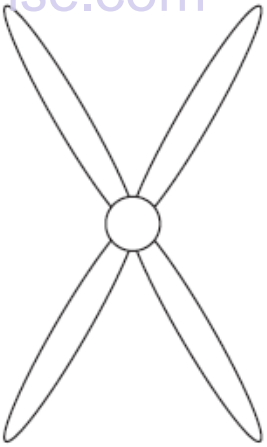
Complete the diagram below to show these chromosomes after crossing over has occurred.

(2)

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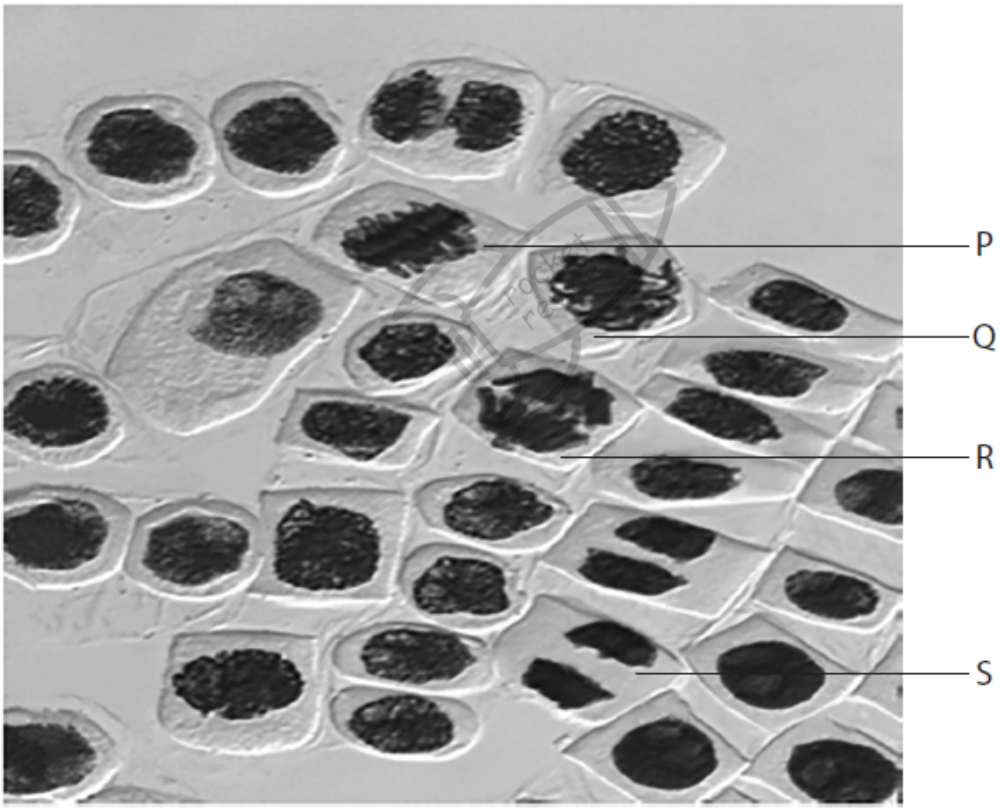




(b) The photograph shows some cells undergoing mitosis.

Each labelled cell is in a different stage of mitosis.

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(i) Which cell is in prophase?

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(1)

☒ **A** P

☒ **B** Q

☒ **C** R

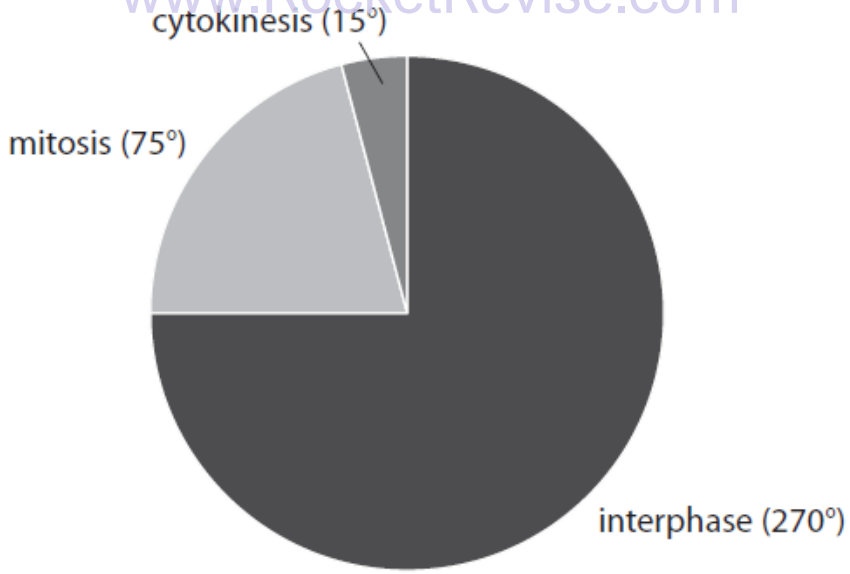
☒ **D** S

(ii) In some embryo cells, interphase can last an average of nine hours.

The diagram shows the relative proportions of time spent in each part of the cell cycle.

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Calculate how long cytokinesis would take in these embryo cells.

(1)

Answer ..... hours

(iii) Calculate the total number of cells resulting from one cell dividing by mitosis eight times.

(1)

Answer .....

(c) Ewing's sarcoma is a type of cancer.

Resveratrol, SRT1720 and Vincristine are drugs used to kill cancer cells.

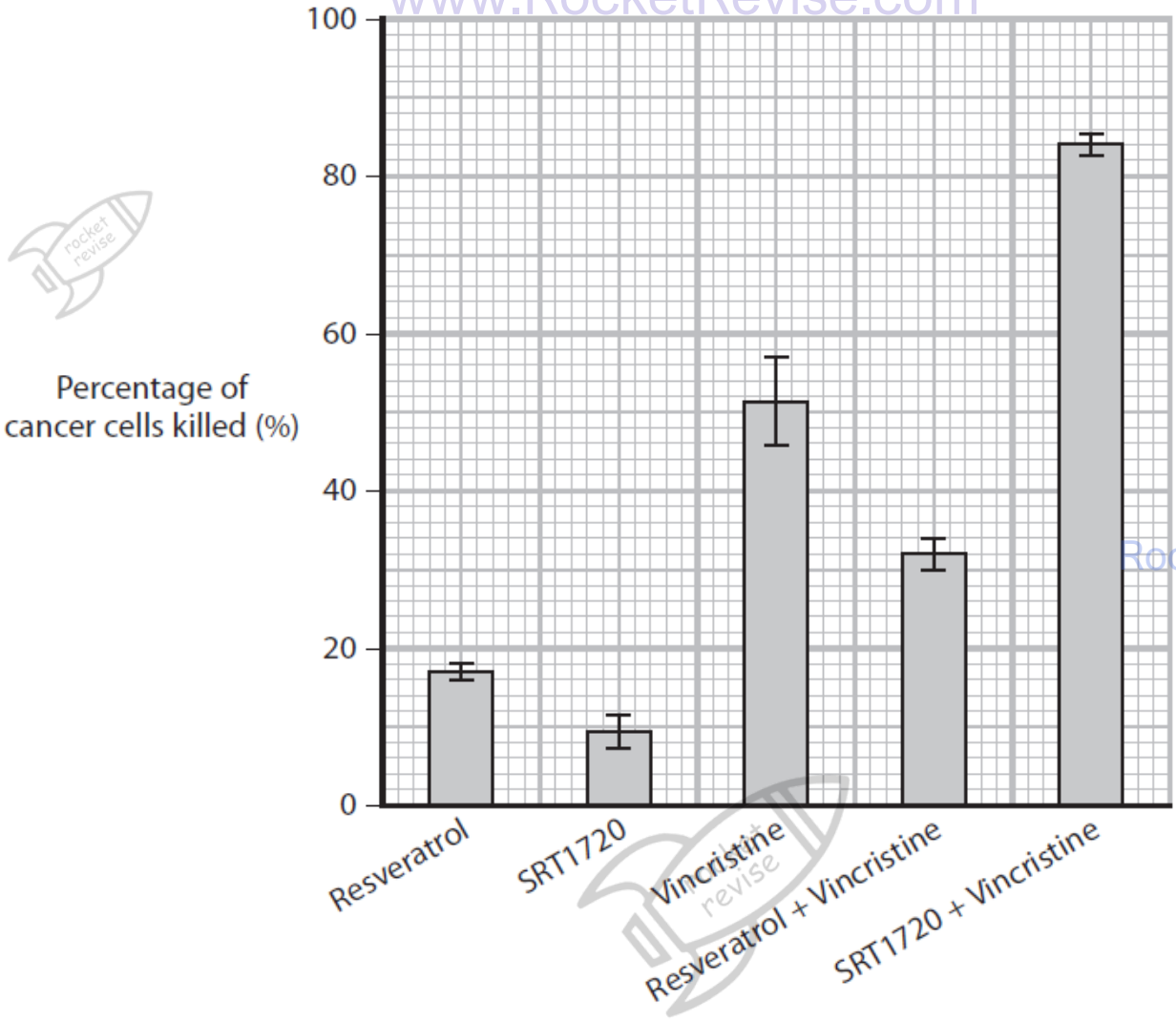
An investigation was carried out to determine the most effective treatment for patients with Ewing's sarcoma.

The graph shows the results of this investigation.

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(i) Vincristine prevents spindle fibres from shortening during mitosis.

This leads to the death of the cell.

At which stage of mitosis do the spindle fibres contract?

(1)

☒ **A** anaphase

☒ **B** metaphase

☒ **C** prophase

☒ **D** telophase

(ii) Evaluate the effectiveness of the three drugs used in this investigation.

Use the information in the graph to support your answer.

(4)

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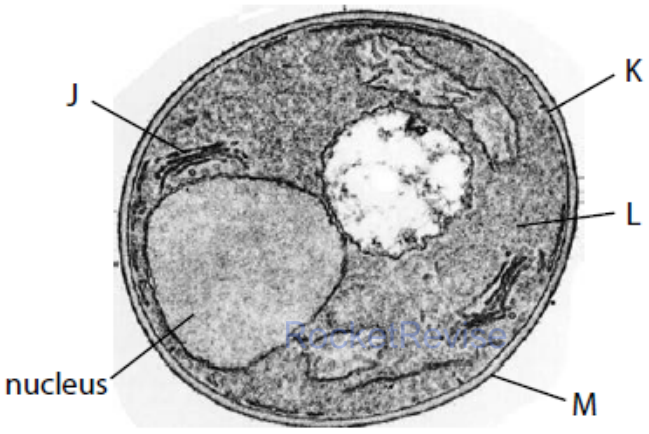
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(Total for question = 10 marks)

Q16.

The photographs show yeast cells, seen using a light microscope and an electron microscope.





Yeast cells seen using a light microscope

Yeast cell seen using an electron microscope

Used under CC License from: https://commons.wikimedia.org/wiki/File:Zygosaccharomyces_bailii_cells.jpg

(a) Which structure identifies yeast as a eukaryotic organism?

(1)

☒ **A** J

☒ **B** K

☒ **C** L

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☒ **D** M

(b) Explain why structure J can be seen using the electron microscope but not the light microscope.

(2)

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(c) Explain why the nuclear envelope cannot be seen as two membranes using this electron microscope.

(2)

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(d) Yeast cells reproduce asexually by a process called budding.

The parent yeast cell produces a bud.

(i) Explain the importance of mitosis in budding.

(3)

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(ii) Once the bud is large enough, it separates from the parent yeast cell.

The rate at which budding happens depends on the availability of oxygen and nutrients.

Suggest why the availability of oxygen and nutrients determines the rate of budding.

(3)

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(Total for question = 11 marks)

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Q17.

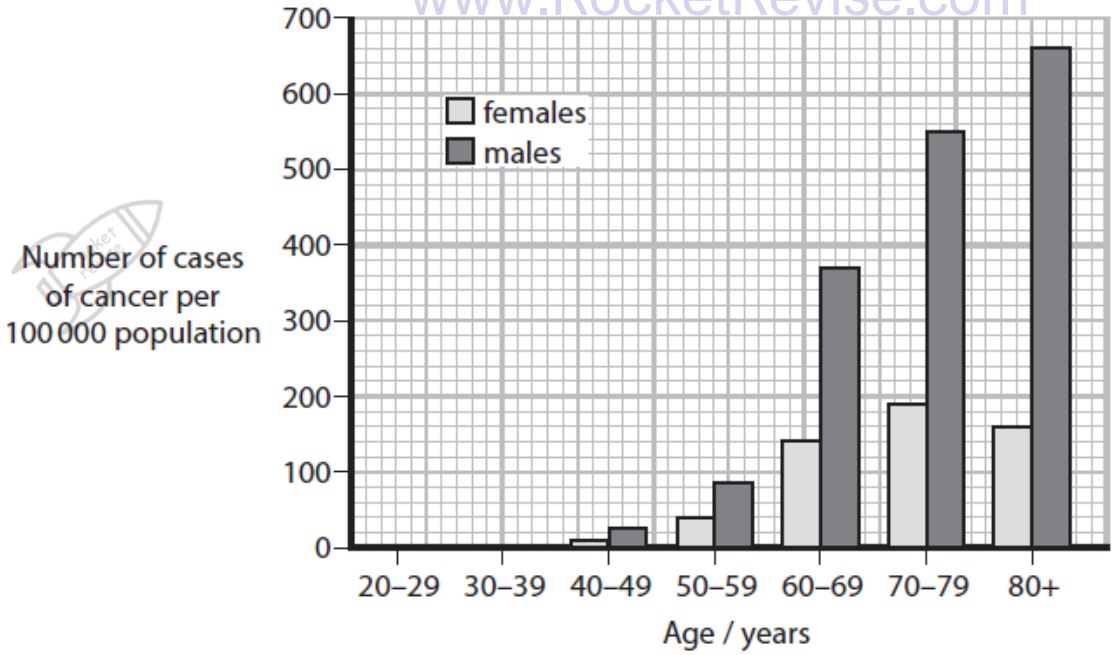
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A change in the base sequence of a gene can give rise to cancer.

The graph shows the number of cases of one type of cancer in a human population.

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(a) (i) Deduce the effect of age and sex on the number of cases of this cancer.

(3)

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(ii) A city had a population of 2 million people. Three per cent (%) of this population were males over the age of 80.

Calculate how many of the males over the age of 80 would have this cancer.

(2)

Answer .....

(b) A sample of tissue was taken from a patient with this type of cancer in order to calculate the mitotic index.

(i) This sample was stained before being viewed using a microscope.

State why the cells in this sample were stained before being viewed using a microscope, in order to calculate the mitotic index.

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(1)

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(ii) Describe how the mitotic index of the tissue sample would be determined.

(2)

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(c) One treatment for this type of cancer prevents the condensing of chromosomes of cancer cells.

(i) In which stage of mitosis do chromosomes start to condense?

(1)

☒ **A** anaphase

☒ **B** metaphase

☒ **C** prophase

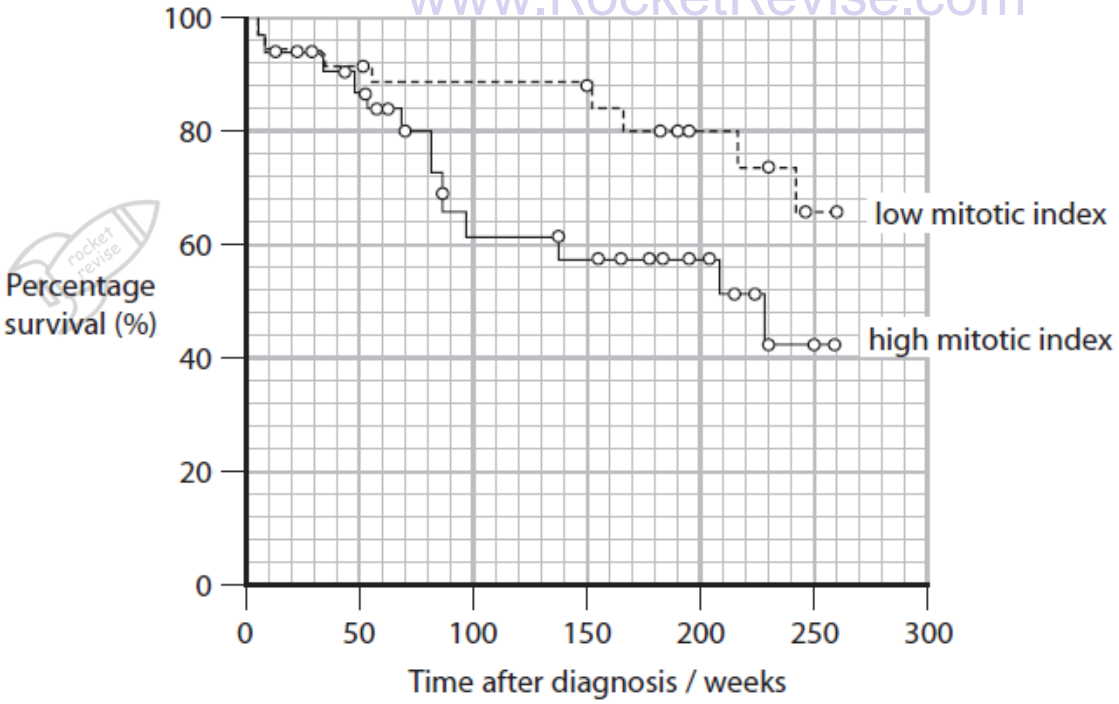
☒ **D** telophase

(ii) The graph shows the effect of the mitotic index on the percentage survival for people with a type of cancer.

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Explain how preventing the condensation of chromosomes would affect the survival of a patient with this type of cancer.

(3)

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(Total for question = 12 marks)

Q18.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer

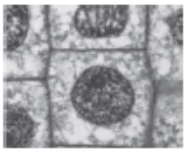
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with a cross ☒.

Interphase and mitosis are parts of the cell cycle.

(a) The photograph shows a plant cell in interphase.



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(i) Interphase is the part of the cell cycle when mitosis is not taking place.

Which of the following processes occur during interphase?

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1. cell growth

2. genetic material is replicated

3. organelles are synthesised

(1)

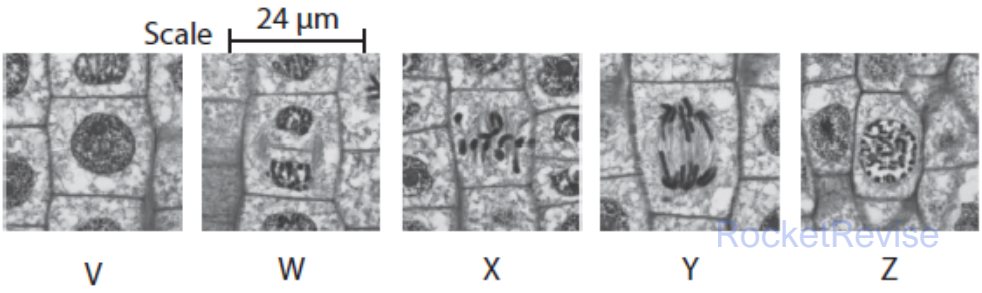
☒ **A** 1 only

☒ **B** 1 and 2 only

☒ **C** 2 and 3 only

☒ **D** 1, 2 and 3

(ii) The photographs show some stages in the cell cycle.



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Which row shows the correct sequence of these stages in the cell cycle?

(1)

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	Sequence of stages				
☒ **A**	V	X	W	Y	Z
☒ **B**	V	X	Y	Z	W
☒ **C**	V	Z	X	Y	W
☒ **D**	V	Z	W	X	Y

(iii) Calculate the magnification of photograph **W**.

Use the scale bar to help you.

(1)

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Answer .....

(b) The mitotic index of a tissue was calculated.

The tissue had 450 cells in mitosis and the mitotic index was 37.5%.

Calculate the number of cells in this tissue that were in **interphase**.

(2)

Answer .....

(c) Describe the events that occur during prophase of mitosis.

(2)

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(d) Some new cells produced by mitosis are used for repairing damaged tissues.

Some new cells produced by mitosis are used for growth of tissues.

(i) Give an example of a tissue that could be produced in a plant.

(1)

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(ii) Give **one role**, other than for growth or repair, for new cells produced by mitosis.

(1)

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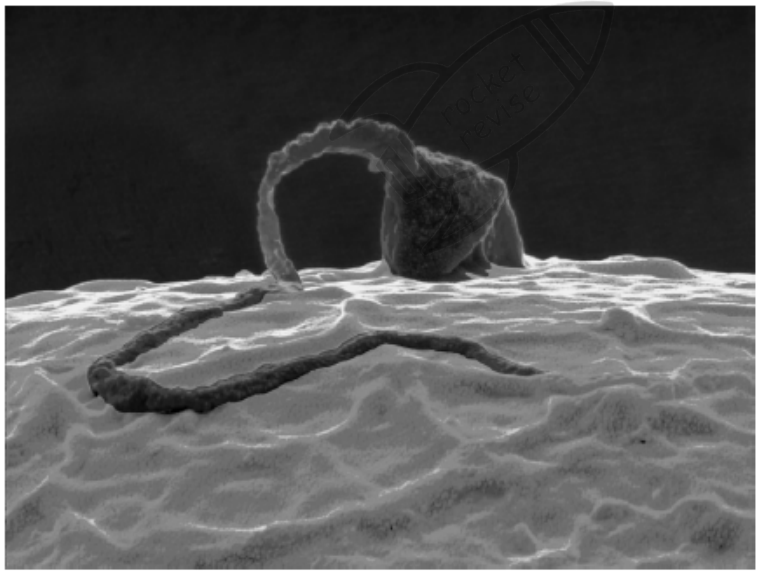
(Total for question = 9 marks)

Q19.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows a sperm cell beginning to enter an egg cell.



(Source: © THIERRY BERROD, MONA LISA PRODUCTION / SCIENCE PHOTO LIBRARY)

(a) A sperm cell swims by rotating its flagellum.

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Which of the following statements about sperm cells are correct?

1. sperm cells are diploid

2. the flagellum rotates using energy

3. sperm cells have a streamlined shape to increase resistance

(1)

☒ **A** 1, 2 and 3

☒ **B** 1 and 2 only

☒

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C 2 and 3 only

☒ **D** 2 only

(b) Describe how the nucleus of a sperm cell enters an egg cell.

(3)

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(c) Cortical granules in the egg cell contain enzymes.

These enzymes prevent more than one sperm entering the egg cell.

(i) Describe the role of the rough endoplasmic reticulum and the Golgi apparatus in the production of these enzymes.

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(3)

(ii) Suggest how these enzymes will prevent more than one sperm nucleus entering the egg cell.

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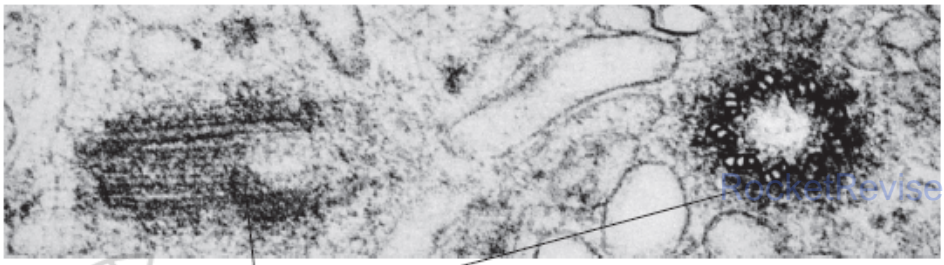
(3)

(Total for question = 10 marks)

Q20.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows two views of an organelle, observed using an electron microscope.



(Source: © Biology Pics / Science Photo Library)

(a) (i) Which is the correct function of this organelle?

(1)

☒ **A** formation of lysosomes

☒ **B** formation of microtubules

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☒ **C** formation of ribosomal RNA

☒ **D** formation of vesicles

(ii) Suggest why the two views of this organelle look different.

(1)

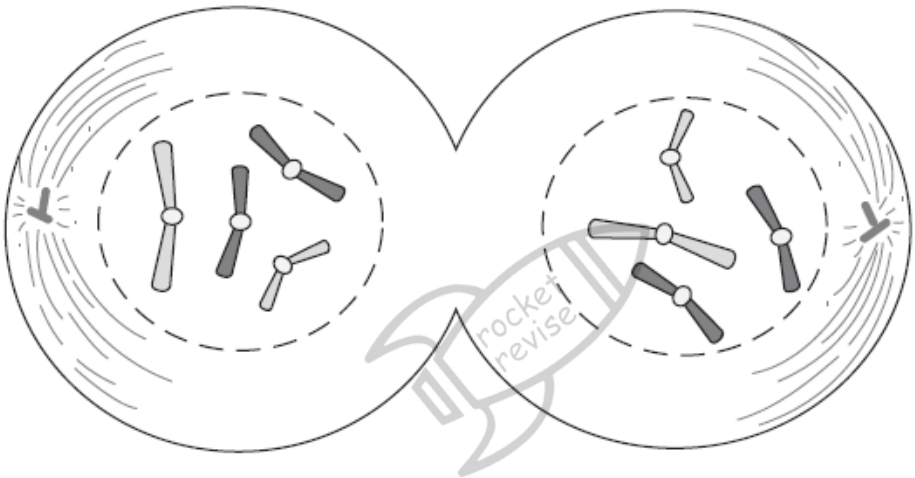
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(b) The diagram shows a cell in telophase of mitosis.

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(Source adapted from: https://www.shutterstock.com/image-vector/telophase-phase-cell-cycle-1678454008)

(i) Draw **this** cell to show it in **metaphase** of mitosis.

(3)

(ii) Describe what happens in **anaphase** of mitosis.

(2)

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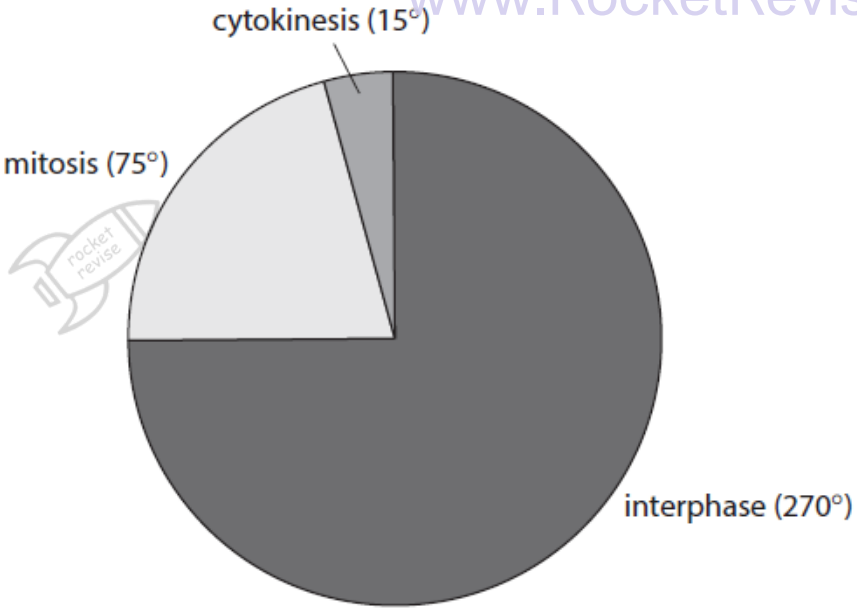
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(c) The diagram shows the relative proportions of the time spent in each stage of the cell cycle.

In these cells, cytokinesis takes 20 minutes.

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Calculate how long interphase would take in these cells.

Give your answer in **hours**.

(1)

..... hours

(d) State how the mitotic index of a tissue could be calculated.

(1)

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(Total for question = 9 marks)

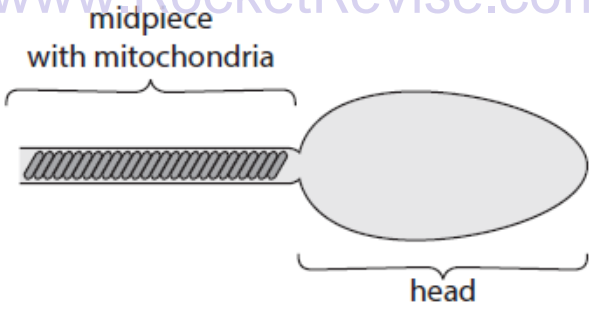
Q21.

A sperm cell is specialised for its function.

(a) The diagram shows part of a labelled sperm cell.

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(i) Complete the diagram by drawing and labelling a flagellum and an acrosome.

(1)

(ii) Give the function of the mitochondria in the midpiece of the sperm.

(1)

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(b) Some male mice will mate with only one female and some will mate with more than one female.

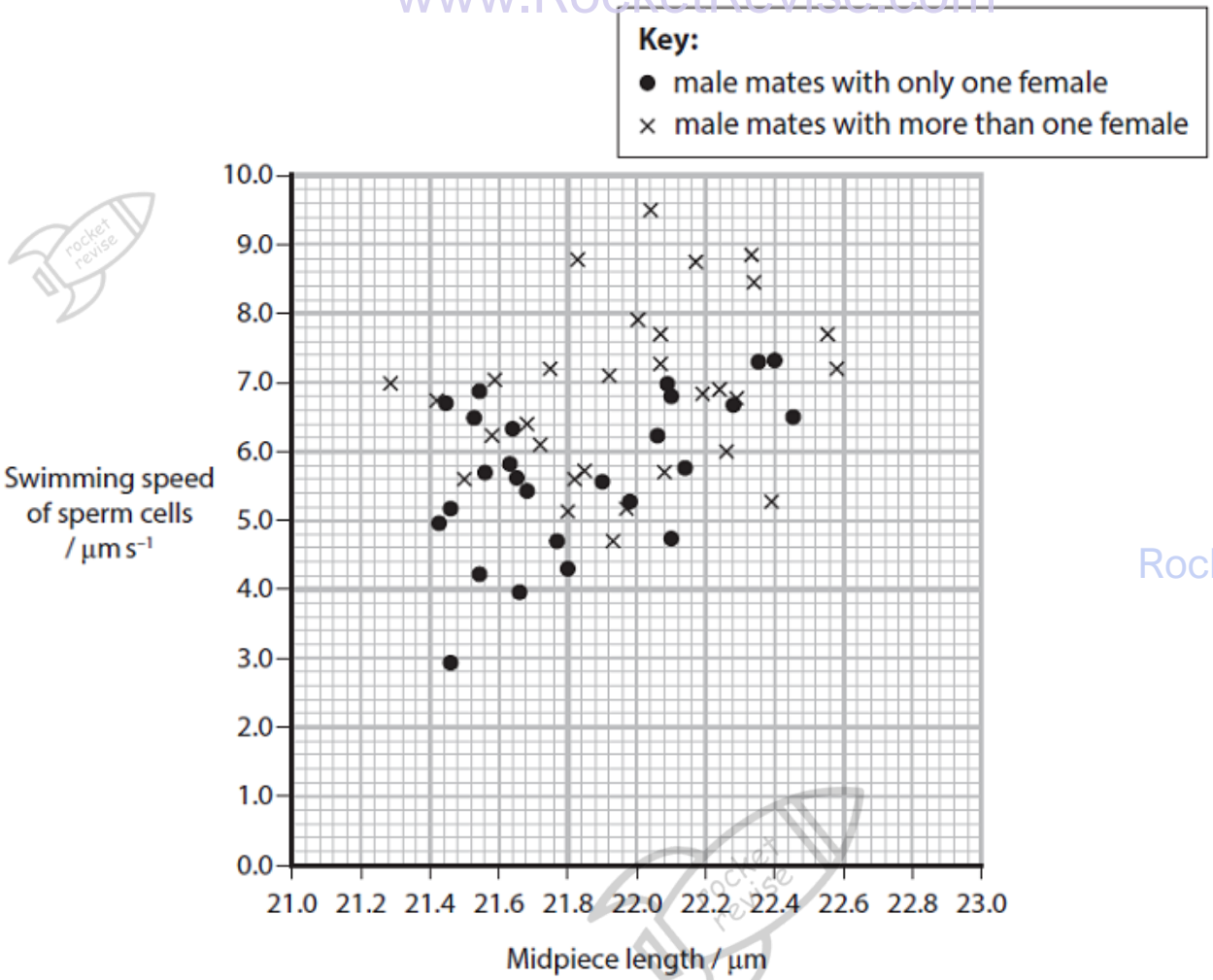
The swimming speed of sperm cells of mice has been investigated.

The graph shows the relationship between midpiece length and swimming speed of sperm cells of mice.

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(i) Give **two** conclusions that can be drawn from these data.

(2)

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2 .....

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(ii) Give the name of a statistical test that could be used to assess the strength of the relationship between length of midpiece and swimming speed of sperm cells.

(1)

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(Total for question = 5 marks)

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